

Mahesh Ponnam

Data Scientist

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SUMMARY

Experienced Data Scientist with 3+ years of experience building and deploying machine learning, NLP, and GenAI solutions for ESG analytics, credit risk, and fraud detection use cases. Strong hands-on experience in Python, SQL, PySpark, AWS, and Snowflake, with a focus on developing reliable data pipelines, predictive models, and model explainability workflows. Skilled in managing the full model lifecycle, supporting real-time analytics, and working with business and engineering teams to deliver measurable improvements in accuracy, efficiency, and risk control.

SKILLS

Programming & Scripting: Python (Pandas, NumPy, scikit-learn, PyTorch), SQL, PySpark

Machine Learning & Statistical Modeling: Supervised Learning, Unsupervised Learning, Regression, Classification, Ensemble Models (Random Forest, Gradient Boosting), Feature Engineering, Feature Selection, Model Validation, A/B Testing

Natural Language Processing & GenAI: Transformer Fundamentals, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, LangGraph, spaCy

MLOps & Model Deployment: AWS (EC2, ECR, SageMaker), Docker, CI/CD (GitHub Actions), Kubernetes, Model Deployment, Model Monitoring, Data & Model Drift Detection, Amazon Bedrock, Agentcore

Data Engineering & ETL: ETL Pipeline Design, Data Cleaning, Data Transformation, Scalable Data Processing

Data Visualization & Explainability: Power BI, Tableau, SHAP Explainability, Model Evaluation & Interpretability, Partial Dependence Plots (PDP), Feature SHAP Analysis

Domain Expertise: Fair Lending Analysis, Credit Risk Assessment, Fraud Detection, Financial Metrics & KPIs

Regulatory & Compliance Knowledge: Regulatory Compliance Frameworks, Basel III, IFRS 9

EXPERIENCE

Northern Trust, USA

Feb 2025 – Present

Data Scientist

- Engineered a unified ESG and factor-data pipeline using Python, Snowflake, and AWS Glue, automating ETL workflows to improve data accuracy by 18% and reduce manual reconciliation effort by 20%.
- Developed and deployed ESG-focused machine learning and NLP models using scikit-learn, PyTorch, AWS SageMaker, accelerating unstructured disclosure processing by 28% while improving ESG signal detection accuracy.
- Designed and implemented real-time ingestion pipelines with AWS Lambda, S3, Redshift, enabling scalable, automated ESG data flows and supporting near real-time analytics across enterprise investment datasets.
- Optimized ESG workflow dashboards and semantic data models in Power BI, integrating LLM-driven insight generation to summarize disclosures and risk drivers, delivering near real-time visibility into portfolio ESG exposures and improving investment decision efficiency by 33%.
- Built and operationalized factor-timing algorithms using market, macroeconomic, and sentiment data, automating portfolio rebalancing recommendations and improving risk-adjusted returns by 25% through predictive analytics.

Mphasis, India

Apr 2020 – Nov 2022

Data Scientist

- Orchestrated design and rollout of Mphasis RiskAI and Mphasis FraudGuard, applying predictive and anomaly detection models for credit risk and fraud analytics, increasing approval accuracy 18% and lowering fraud 15%.
- Constructed scalable ETL pipelines using PySpark, SQL, and AWS Glue to process large-scale financial data, enforcing validation, cleansing, and feature engineering to ensure consistent data quality across platforms.
- Derived advanced features from transactional, demographic, and macroeconomic datasets, implementing credit utilization and debt-to-income metrics that elevated model precision 12% across supervised and unsupervised learning workflows in production.
- Provisioned scalable ML models and APIs via AWS Lambda and EC2, integrating real-time streams, CloudWatch monitoring, and CI/CD with Docker to accelerate decisions 20% and support continuous validation.
- Applied AI explainability using SHAP, LIME, and Captum with Tableau dashboards, optimizing PySpark through partitioning and caching, and containerizing workflows with Docker to reduce false positives by 10–12% while meeting Basel III and IFRS 9 compliance.

EDUCATION

Masters of Science in Computer and Information Systems Security

Dec 2024

Wilmington University, USA

Bachelor of Technology in Computer Science and Engineering

Jul 2022

Vignana Bharathi Institute of Technology, India